

Improved MPC for PMSM drives to Reduce Switching Losses

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Abstract— This paper presents an improved model predictive control (MPC) for permanent magnet synchronous motor (PMSM) fed by a voltage source inverter (VSI), in order to reduce switching losses. The MPC strategy is based on a cost function generally established for performance purposes and leads to switching states to be applied for VSI control. In this work, the cost function is improved in order to reduce changing states so to reduce switching losses. Simulations are carried out to illustrate the theoretical development and results demonstrate the effectiveness of the proposed approach.

Keywords—PMSM; MPC; Cost function, Switching losses.

I. INTRODUCTION

Recently, permanent magnet synchronous motors (PMSM) drives have been widely used in many industrial applications, like electrical vehicles, aircraft, nuclear power stations, submarines, robotic applications, medical and industrial servo drives. Indeed, they are preferred in many applications for their high efficiency, high torque-to-inertia ratio, high power factor, faster response and rugged construction [1].

Several control schemes have been proposed for the PMSM. Their performances are strongly related to VSI power switches changing states strategy which depends on control scheme. The commonly used are field-oriented control (FOC) and direct torque control (DTC) [2, 3]. Other control schemes are also widely used like those based on hysteresis and deadbeat-based controllers [4, 5].

Recently, model predictive control (MPC) has attracted much intention from the researchers for high power applications due to several advantages: good dynamic response, straightforward inclusion of non-linearities. MPC approach requires a system model over a horizon in time. MPC design is performed by solving an optimization problem providing the new states to be applied in future, so, by minimizing a predefined cost function which represents the desired behavior. MPC scheme has been used for the control in a three-phase inverter, matrix converters, rectifiers, rotating devices (induction and synchronous machines), and more [6-9], achieving a simple and flexible control with good performance.

In our application, the MPC should ensure current control and generate VSI switching states. For a two level inverter, only seven states could be reached with eight combinations of

control signal values. This makes easy to solve the optimization problem by directly evaluating all possible switching states.

In this work, authors propose an improved approach to reduce the switching losses by reducing the number of switches that change their states during the sample time interval [10-12]. Switching losses optimization problem have been treated by adding a weight factor to switches changing states number [13, 14]. The idea in this paper is to not add this supplementary weight factor but to keep MPC algorithm standard cost function, and to apply new switch states values only after a decision process, taking into account switching losses, so number of required switching.

In this paper, PMSM and power converter models are presented in section 2. MPC strategy and the proposed approach of switching losses optimization are developed in section 3. Simulation results under Matlab/SIMULINK are shown in section 4. Finally, the contributions of this work are resumed in a conclusion highlighting the advantages of the proposed method.

II. MATHEMATICAL DESCRIPTION OF THE SYSTEM

A. PMSM Model

To simplify the analysis, the model of PMSM is established under the following assumptions. The magnetic saturation is neglected, the back electromagnetic force is supposed sinusoidal and the eddy current, magnetic hysteresis losses and the clogging torque are very small and then neglected. According to the theory in [15], by using the concept of "active flux", any salient-pole rotor machines can be equivalently described by a non-salient-pole rotor machines.

Conventionally, vector control of synchronous machine is developed in terms of (dq) rotatory frame, which is oriented with the rotor magnetic field axis. Using this frame of reference has the advantage of transforming sinusoidal variables into DC values. This yields a model that is similar to that a DC machine and, hence, to a conceptually simple drive implementation. In this frame, each current component of the machine has a clear physical meaning: i_q is proportional to the torque, while i_d is proportional to reactive power [16]. By neglecting the homopolar component, the stator voltages expressed by:

$$\begin{cases} v_d = R.i_d + L_d \frac{di_d}{dt} - \psi_q.\omega \\ v_q = R.i_q + L_q \frac{di_q}{dt} + \psi_d.\omega \end{cases} \quad (1)$$

Where, the stator fluxes given by:

$$\begin{cases} \psi_d = L_d i_d + \psi_r \\ \psi_q = L_q i_q \end{cases} \quad (2)$$

Moreover, the electromagnetic torque expressed as:

$$T_e = \frac{3}{2} p \{ (L_d - L_q) i_d i_q + \psi_r i_q \} \quad (3)$$

As the used PMSM is a non-salient machine, all the inductances are equal: $L_d = L_q = L$. So, (3) simplified as:

$$T_e = \frac{3}{2} p \psi_r i_q \quad (4)$$

Finally, the all-mechanical equations expressed as follows:

$$J \frac{d\Omega}{dt} = T_e - T_L - f\Omega \quad \text{and} \quad \frac{d\theta}{dt} = \omega = p\Omega \quad (5)$$

Where: u_d, u_q are voltage components; i_d, i_q motor phases currents; ψ_d, ψ_q and ψ_r flux in dq -axis respectively and the flux linkage due to permanent-magnets. T_e and T_L are the electromagnetic torque and the loaded torque; f is the viscous friction coefficient and J is the moment of inertia. Finally, θ, Ω, ω and p are respectively the electrical angular, the mechanical speed, the angular frequency and the number of pole pairs.

B. Power Inverter Model

A conventional two level voltage source inverter (VSI) drives the motor. A scheme of this topology and the connection to the motor is presented in figure 1.

The different output voltage values generated by this inverter are determined by the states of the switches in each of its phases: S_a, S_b and S_c . Each of these switching functions can adopt values $S_x \in \{1, 0\}$, where $x \in \{a, b, c\}$.

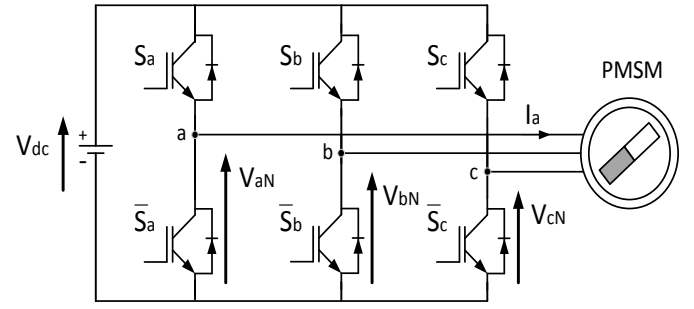


Fig. 1. PMSM fed by the three-phase inverter.

The switching state of the inverter can be summarized by the following equations system:

$$\begin{cases} V_{aN} = \frac{V_{dc}}{3} (2.S_a - S_b - S_c) \\ V_{bN} = \frac{V_{dc}}{3} (2.S_b - S_a - S_c) \\ V_{cN} = \frac{V_{dc}}{3} (2.S_c - S_b - S_a) \end{cases} \quad (6)$$

Where V_{dc} is the DC-Link voltage on fig. 1. Considering all the possible combinations of switching functions (S_a, S_b, S_c), seven distinct voltage vectors are obtained. These are shown in fig. 2. These constitute the finite alphabet for the actuation and correspond to the voltage vectors that can be applied to the machine.

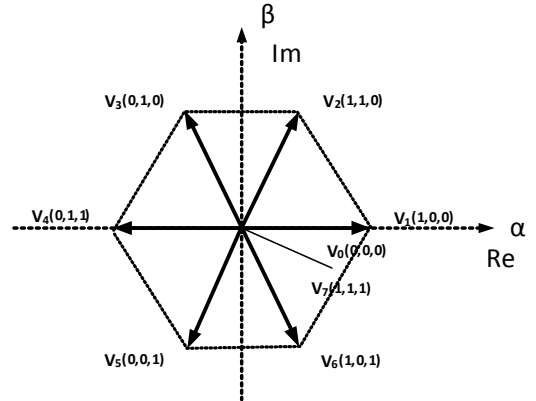


Fig. 2. PMSM fed by the three-phase inverter.

This notation allows one to characterize the output voltage vector as:

$$V_x = V_\alpha + jV_\beta \quad \text{with} \quad x \in \{0, 1, \dots, 7\} \quad (7)$$

III. MODEL PREDICTIVE CONTROL (MPC) SCHEME FOR THE PMSM

A. The Control Strategy

The classical control schemes for the PMSM use a cascaded structure, with an external current or field control.

The proposed scheme preserves this structure, replacing the inner control (PI for FOC, and hysteresis for DTC) with a predictive current controller. This idea is depicted in the drives diagram shown in fig. 3.

The predictive control algorithm can be understood in three main stages:

- i. The first stage consists in the prediction of the motor state for every voltage vector available in the inverter. The predicted state is calculated using the discrete model of the drive.
- ii. The second stage consists in the evaluation of a cost function for every predicted state. This cost function is a representation of the control goals.
- iii. Finally, the switching state of the inverter is adjusted to apply the selected voltage vector to the motor during the next sampling time.

B. Discrete-Time Model

From (1), the state space stator currents of PMSM designed as reference model is given by $\dot{x} = Ax + Bu + D$, where:

$$x = \begin{bmatrix} i_d \\ i_q \end{bmatrix}; A = \begin{bmatrix} -\frac{R}{L} & \omega \\ -\omega & -\frac{R}{L} \end{bmatrix}; B = \begin{bmatrix} \frac{1}{L} & 0 \\ 0 & \frac{1}{L} \end{bmatrix} \quad (8)$$

$$u = \begin{bmatrix} v_d & v_q \end{bmatrix}^T; D = \begin{bmatrix} 0 & -\frac{\psi_r}{L} \omega \end{bmatrix}^T$$

According to (8), the stator current at the next control period can be predicted using the Euler approximation for a sampling time T_s as follow: $\frac{di}{dt} \approx \frac{i(k+1) - i(k)}{T_s}$. Then, the predicted stator currents are expressed by the following system:

$$\begin{cases} i_d(k+1) = (1 - \frac{RT_s}{L})i_d(k) + T_s \omega i_q(k) + \frac{T_s}{L} v_d(k) \\ i_q(k+1) = (1 - \frac{RT_s}{L})i_q(k) - T_s \omega i_d(k) - \psi_r \omega T_s + \frac{T_s}{L} v_q(k) \end{cases} \quad (9)$$

Where $i_d(k+1)$ and $i_q(k+1)$ (respectively $i_d(k)$ and $i_q(k)$) are the predicted d and q stator current components at the $(k+1)^{th}$ sampling time (respectively d and q stator current components during k^{th} sampling time). During each sampling time, the evolution of the d and q stator current components depends on the applied stator voltage components $v_d(k)$ and $v_q(k)$ at the k^{th} sampling time. The voltage vectors generated by the inverter are fixed in the stationary reference frame, but they are rotating vectors in the dq reference frame can be determined through the application of a rotation operation to the $\alpha\beta$ components of the stator voltage vector as shown in equations (10) and table I.

$$\begin{bmatrix} v_d(k) \\ v_q(k) \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} v_\alpha(k) \\ v_\beta(k) \end{bmatrix} \quad (10)$$

TABLE I. SWITCHING STATES AND VOLTAGE VECTORS

S_a	S_b	S_c	V_α	V_β	$V_x = V_\alpha + jV_\beta$
0	0	0	0	0	V_0
1	0	0	$2V_{dc}/3$	0	V_1
1	1	0	$V_{dc}/3$	$V_{dc}/\sqrt{3}$	V_2
0	1	0	$-V_{dc}/3$	$V_{dc}/\sqrt{3}$	V_3
0	1	1	$-2V_{dc}/3$	0	V_4
0	0	1	$-V_{dc}/3$	$-V_{dc}/\sqrt{3}$	V_5
1	0	1	$V_{dc}/3$	$-V_{dc}/\sqrt{3}$	V_6
1	1	1	0	0	V_7

C. Cost Function

A non-negative function that evaluates the distance of the predicted system state to the desired values given by the reference and additional control objectives, has to be devised. Such a function is referred to as a cost function, and assumes a minimum value for the predicted state that best agree with the control objectives [17-19].

In this case, the control objectives are:

- i. Minimization of the torque by ampere ratio, minimization of the i_d current component.
- ii. Tracking of i_q current reference.
- iii. Limitation of the stator current amplitude, maintaining the currents limited.

The cost function designed to achieve this objectives is given by:

$$g = [i_d^* - i_d(k+1)]^2 + [i_q^* - i_q(k+1)]^2 + X_{lim} [i_d(k+1), i_q(k+1)] \quad (11)$$

$$X_{lim} [i_d(k+1), i_q(k+1)] = \begin{cases} \infty & \text{if } |i_d(k+1)| > i_{d_{max}} \text{ or } |i_q(k+1)| > i_{q_{max}} \\ 0 & \text{if } |i_d(k+1)| \leq i_{d_{max}} \text{ and } |i_q(k+1)| \leq i_{q_{max}} \end{cases} \quad (12)$$

Where i_q^* is the quadrature current reference. The first term of the cost function weights the magnitude of the i_d current component. This term penalizes the application of voltages vectors that result in currents that do not generate torque. The second term weights the error between the i_q current component and its reference, which is given by the external speed controller. This term penalizes the application of voltages vectors that take the quadrature current away from its reference.

The last term of the cost function is a nonlinear function for limiting the amplitude of the stator currents, which is defined in (12). Where $i_{d,qmax}$ is the value of the maximum allowed stator current magnitude. In this way, if a given voltage vector generates predicted currents with a magnitude higher than $i_{d,qmax}$ then the cost function will be $g=\infty$, and, in consequence, this voltage vector will not be selected. On the other hand, if the predicted stator currents are below the limits, the cost

function will be composed of the first two terms only and the voltage vector that minimizes the current error will be selected.

Finally, an optimization procedure is applied. This procedure consists in selecting the optimal switching states combination $S_{(a,b,c)}^{opt}$ that leads to the minimal cost function $\min(g^j)_{j \in \{0, \dots, 7\}}$. Then, the voltage vector that minimizes the current error will be selected and will be applied to three legs of the inverter to run the machine in optimal conditions.

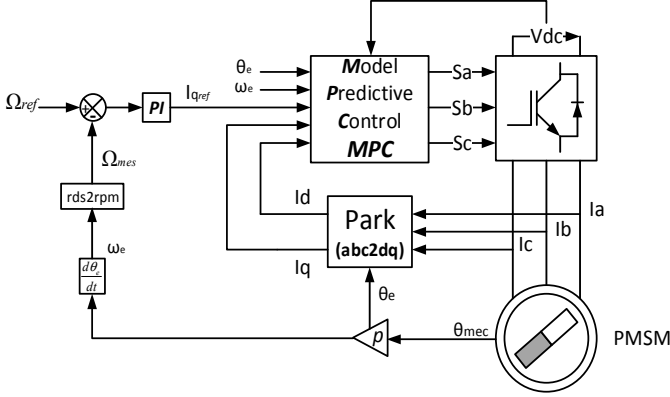


Fig. 3. MPC based control for the PMSM.

D. Minimization of the switching losses

As in power converters, one of the major measures of control effort is the switching frequency. It is important in many applications to be able to control or limit the number of commutations of the power switches. On the other hand, the switching losses depend also the voltage and current values at the moment of switching. A simple model of the switching process can be considered for direct minimization of the switching losses using predictive control.

To directly consider the reduction in the number of commutations in the cost function, a simple approach is to include a term in it that covers the number of switches that change when the switching state $S(k)$ is applied, with respect to the previously applied switching state $S(k-1)$.

The optimized cost function can be described as bellow:

$$g_{opt} = g + Y_{lim}(\lambda_{S_x}) \quad (13)$$

Where λ_{S_x} is the number of switches that change states when the switching state $S(k)$ is applied. As S the switching state vector is defined in (14) and presents the switching state of each inverter leg:

$$S = [S_a, S_b, S_c]^T \quad (14)$$

Where each element S_x represents the state of a switch and has only two states, one or zero, then the number of switches that change from $S(k-1)$ to $S(k)$ is:

$$\lambda_{S_x} = \sum_x |S_x(k) - S_x(k-1)| \quad \text{with} \quad x \in \{a, b, c\} \quad (15)$$

By using $Y_{lim}(\lambda_{S_x})$ defined in (16), certain switching states can be canceled from the minimization process without affect the commutation process itself, but predicts the losses that a given switching state would produce, penalizing states that imply higher losses.

$$Y_{lim}(\lambda_{S_x}) = \begin{cases} \infty & \text{if } \lambda_{S_x} > 1 \\ 0 & \text{if } \lambda_{S_x} \leq 1 \end{cases} \quad (16)$$

For example, if the current used voltage vector is $V_5(0,0,1)$ and the next optimal voltage vector is $V_2(1,1,0)$; then in this case three switches of inverter legs change their states and $\lambda_{S_x} = 3$. In this way and by adding the term of optimization; the cost function will be $g_{opt} = \infty$ and in consequence this voltage vector will not be selected. On the other hand, the optimal voltage vector from $V_4(0,1,1)$, $V_6(1,0,1)$ or $V_0(0,0,0)$ will be selected when it be noted that only one switch changes its state.

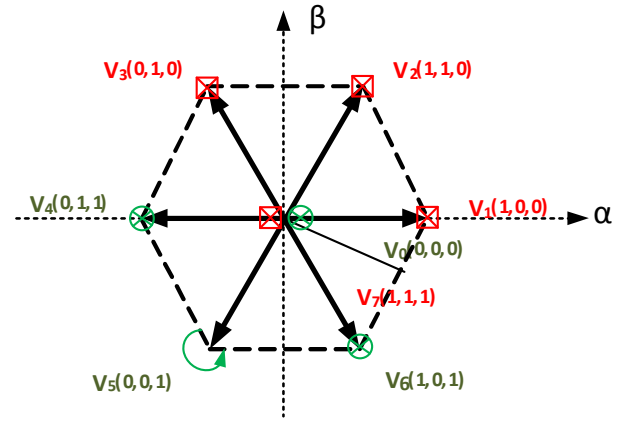


Fig. 4. Voltage vectors can be selected and canceled.

Fig. 4 shows the last example of the choice of the optimal voltage vectors that can be selected and there can be canceled despite that some one of them can be the optimal voltage vector. One among the voltage vectors marked with green circle can be selected and there marked with the red square cannot be selected. On this way the switching frequency is minimized. There are one switch changing it state between the actual vector and the next vector.

IV. SIMULATION RESULTS

A. Standard cost function

The performance of the proposed control scheme has been tested through simulations using Matlab/Simulink models. The control algorithm was implemented in C to favor portability. Simulation results of PMSM electrical and mechanical parameters using the model predictive control scheme are shown in figures (6) and (7). In this part of simulation results, the used cost function of the MPC is described in (11).

All simulation results are obtained using the motor parameters listed in Table II and the sample time chosen equal to $T_s=10\mu s$. It should be noted that during simulation procedure, the following operation conditions were taken into account:

- The dc-link voltage equal to 560V.
- The d axis stator current reference i_d^* was set to zero in order to obtain the maximum torque at the minimum current.
- The q axis stator current reference i_q^* was obtained from the external PI controller that used for speed control and generates the reference for the torque-producing i_q^* .

Fig. 5 presents the test bench of the PMSM running where the machine starts with 300rpm speed reference then a speed reference step change to 600rpm (nominal speed) is performed at time $t=2s$, finally a speed reversal to -500rpm at $t=4s$ is applied. For another hand, a load step of nominal torque value (832 N.m) is applied at times 1s and 3s.

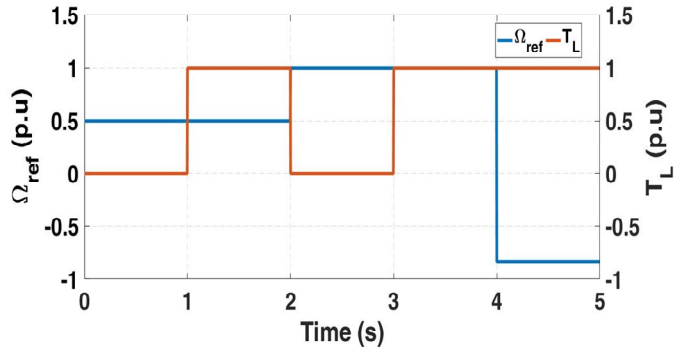
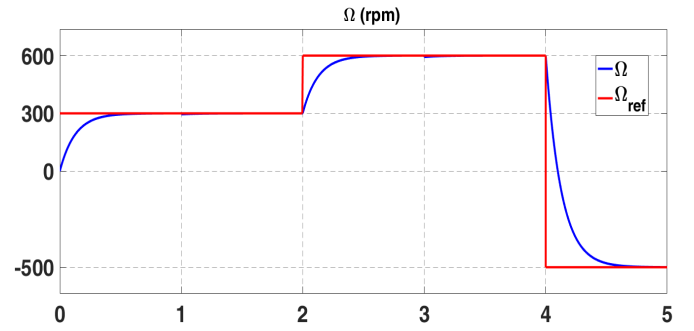
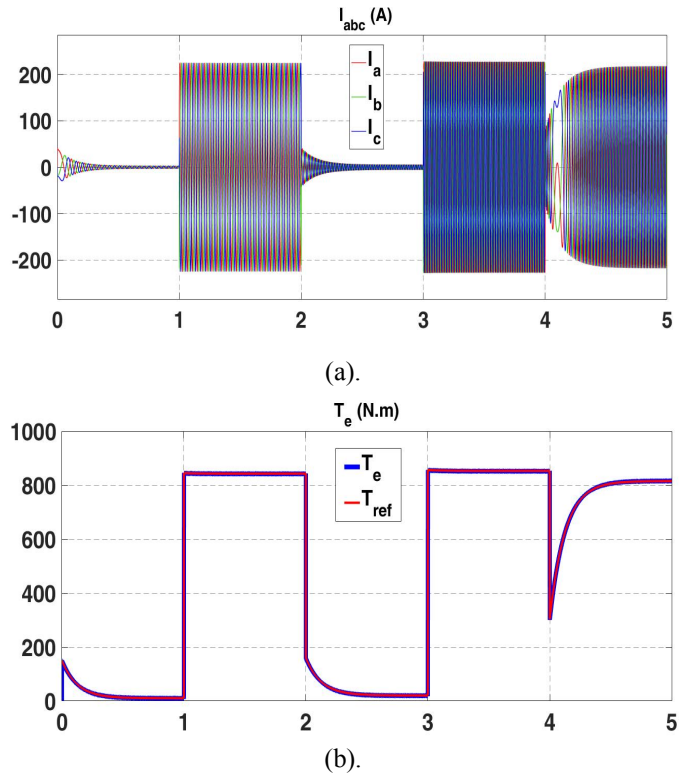
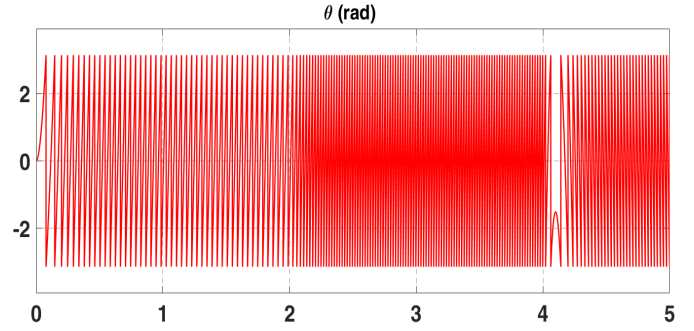


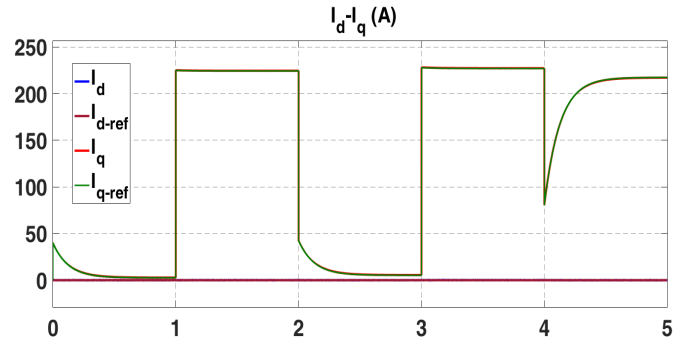
Fig. 5. Speed reference and torque load applied to PMSM.



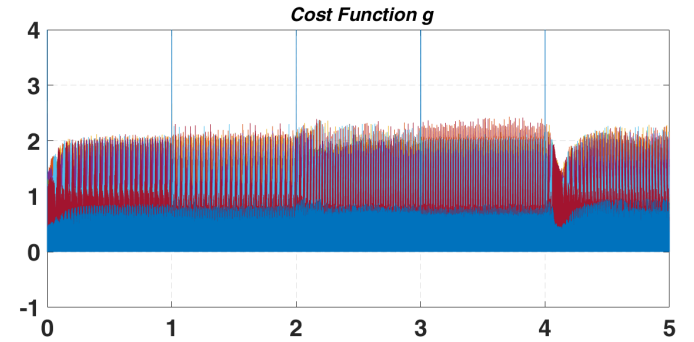
(c).



(d).



(e).



(f).

Fig. 6. (a). PMSM stator currents waveforms. (b). PMSM electromagnetic torque. (c). PMSM rotor speed. (d). PMSM rotor position. (e). PMSM stator currents in rotating frame I_{d-q} . (f). Cost function.

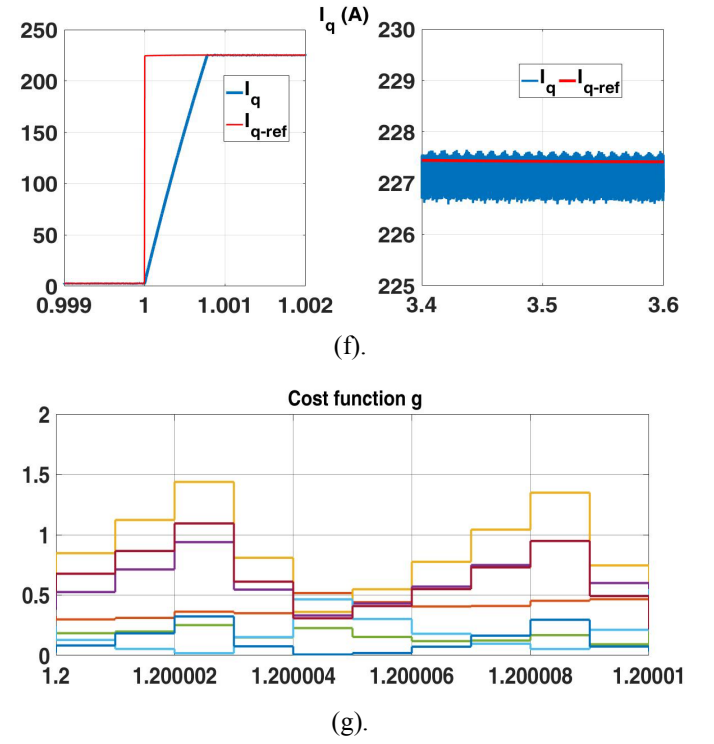
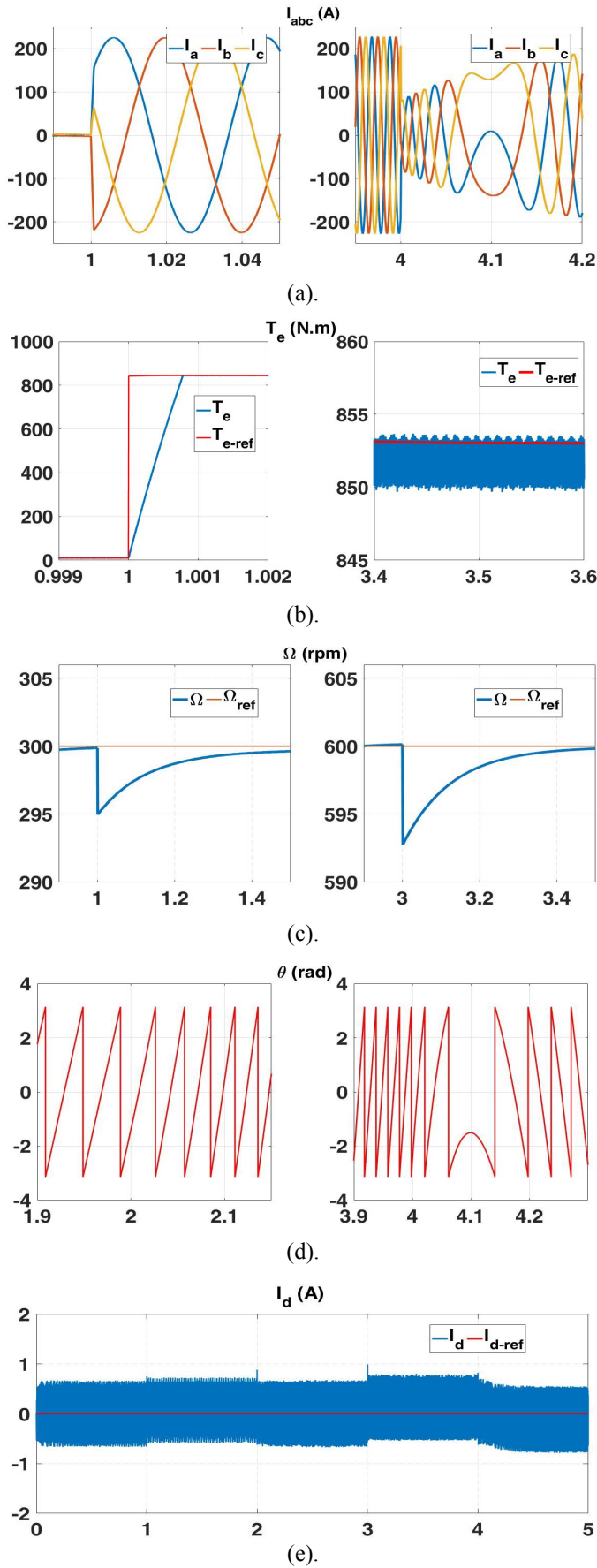


Fig. 7. (a). PMSM stator currents response to a step reference of 832 N.m applied to the load torque and to change of rotor speed from 300rpm to -500rpm. (b). PMSM electromagnetic torque response to a step reference of 832 N.m applied to the load torque and its oscillation compared to the reference. (c). PMSM rotor speed response to a step reference of 832 N.m applied to the load torque. (d). PMSM rotor position response to change of rotor speed from 300rpm to 600rpm and from 600rpm to -500rpm. (e). PMSM stator current I_d evolution compared to its reference. (f). PMSM stator current I_q response to a step reference of 832 N.m applied to the load torque and its oscillation compared to the reference. (g). Zoom of cost function evolution.

Fig.6 (a) and 7 (a). show respectively the behavior of the PMSM stator currents regarding the external perturbation (load and speed change). According to these results, it is clear that the stator currents i_{abc} increases quickly (about 1ms) when the load torque increases. Fig.6 (c) and 7 (c) show respectively the behavior of the PMSM rotor speed regarding the load changing and compared with its reference. There is a slight speed fall when the external load is applied and then the motor speed returns to its normal value quickly, exhibiting strong robustness against external load disturbance. During this test condition, the PI-based speed controller compensates this disturbance. Hence, used speed controller achieves efficient speed control with good dynamic and accuracy.

It can be seen from these results that all objectives of the control are achieved during the tests. Fast tracking of the torque-producing current i_q is achieved while the i_d current is near zero as it is shown in fig.6 (e). Where we can see in fig.7 (b), that the MPC presents lower torque ripple (about 4N.m compared to its reference) and the same note in fig. 7 (e) and (f) that presents a less current harmonics. All these results validate the good dynamic performance of the proposed method.

B. Optimized cost function

In this part of simulation results, the optimized cost function g_{opt} that described in (13), (15) and (16) is used. The comparative results between some PMSM electrical and mechanical parameters that controlled in the first time with the standard MPC scheme and in the second time with the optimized MPC scheme are presented in the following figures.

Fig. 8 presents the power switch keys evolution where we note that fig. 8 (a).shows higher switching frequency contrary of the fig. 8 (b). that shows a lower switching frequency. Moreover, the importance of use the optimized cost function of the MPC scheme is the reduction of switching states number of power switches in the optimization problem. Then, the optimized MPC allows to decrease the use of switches in the power inverter and subsequently to optimize the power losses.

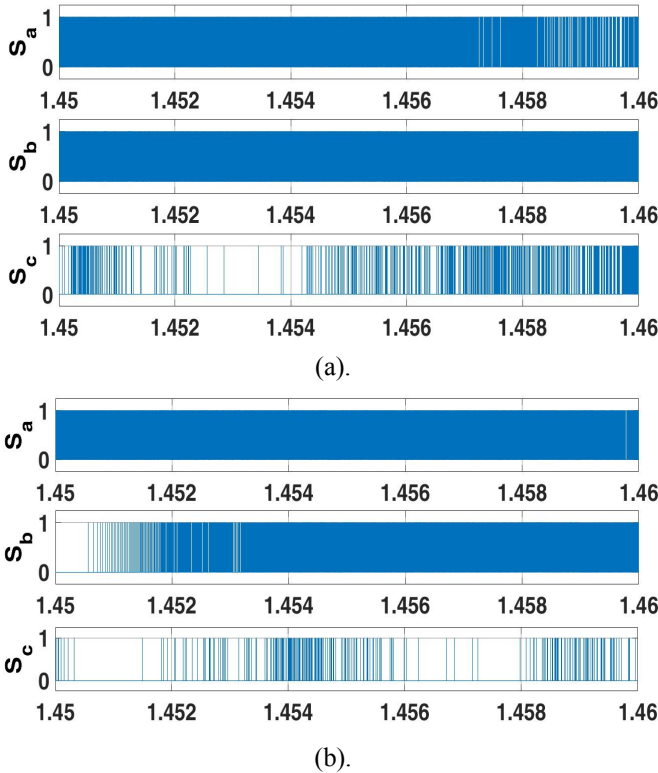


Fig. 8. (a). Power switch keys with standard cost function . (b). Power switch keys with optimized cost function.

V. CONCLUSION

In first part of this paper, the model predictive control scheme of PMSM drive has been presented, which does not make use of any linear controller or any type of modulator and could be competitive to the well-known controls as: Field Oriented Control (FOC), Direct Torque Control (DTC), etc.... Where, the speed controller is performed via a PI controller that generates the i_q current reference component. The main idea of MPC is based on an internal current control loop. All simulation results validate the effectiveness of the proposed method. It is be concluded that the proposed MPC can improve steady state performance with excellent dynamic performance.

In second part, to reduce the number of power switches used in the MPC algorithm, a simple method that reduces the number of switches used each sampling time has been proposed. Comparative results show that reducing the number of used switches, not affected the good control performance and the goal of switching losses optimization is obtained. Therefore, future work will focus on the energy losses optimization in VSI to increase the performance of the system.

TABLE II. PMSM PARAMETERS

Variable	Rating Values
Rated Power (P_n)	53 kW
Rated Torque (T_e)	832 N.m
Rated Current (I_n)	122 A
Flux Linkage (Ψ_r)	0.5 Wb
Pole Pair (p)	5
Rated Speed (Ω)	600 rpm
Armature Resistance (R)	87 mΩ
dq-axis Inductance ($L_{sdq}=L$)	0.8 mH
Rotor Inertia (J)	0.6 Kg.m ²
Viscous Friction Coefficient (f)	0.3129 N.m.s/rad

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REFERENCES

- [1] I. Maaoui-Ben Hassine, M.W Naouar and N. Mrabet-Bellaaj, "Predictive Control Strategies for Wind Turbine System Based on Permanent Magnet Synchronous Generator", ISA Transactions, Elsevier, Vol. 62, pp: 73-80, Mai 2016.
- [2] M. Bouragaoui, H. Ben Attia-Sethom, I. Slama Belkhdja, "Real-time encoder faults detection and rotor position estimation for permanent magnet synchronous motor drives fault tolerant sensorless control using digital signal controller", Mathematics and Computers in Simulation, In Press, 3 Oct 2015.
- [3] M. Abassi, A. Khlaief, O. Saadaoui, A. Chaari and M. Boussak, "Performance Analysis of FOC and DTC for PMSM Drives using SVPWM Technique", 16th International Conference on Sciences and Techniques of Automatic Control and Computer Engineering-STA'15, Monastir Tunisia, December 21-23, 2015.
- [4] Hongyu Tang, W. Zhao, "Deadbeat Control of Permanent Magnet Synchronous Motor Based on Sliding Active Disturbance Rejection Controller", 18th International Conference on Electrical Machines and Systems (ICEMS), Oct. 25-28, 2015, Pattaya City, Thailand.
- [5] Purnima Purohit, Menka Dubey, "Analysis and Design of Hysteresis Current Controlled Multilevel Inverter Fed PMSM Drive", 2014 IEEE Students Conference on Electrical, Electronics and Computer Science.
- [6] Vaclav Šmidl, Vaclav Macha, Štěpán Janouš, Zdeněk Peroutka: "Analysis of cost functions and setpoints for predictive speed control of PMSM drives ", 18th European Conference on Power Electronics and Applications (EPE'16 ECCE Europe), pp: 1-8, 5-9 Sept. 2016.
- [7] Yongchang Zhang, Suyu Gao, Wei Xu: " An Improved Model Predictive Current Control of Permanent Magnet Synchronous Motor

- Drives ", 2016 IEEE Applied Power Electronics Conference and Exposition (APEC), pp: 2868-2874, 20-24 March 2016.
- [8] Esteban J. Fuentes, Jose Rodriguez, César Silva, Sergio Dfaz, Daniel E. Quevedo: " Speed Control of a Permanent Magnet Synchronous Motor Using Predictive Current Control ", IEEE 6th International Power Electronics and Motion Control Conference (IPEMC'09), pp: 53-58, 17-20 May 2009.
 - [9] I. Maaoui-Ben Hassine, M. W. Naouar, N. Mrabet-Bellaaj: "Model Predictive-Sliding Mode Control for Three-Phase Grid-Connected Converters", IEEE Transactions on Industrial Electronics, Vol. 64, Issue:2, pp: 1341-1349, 2017.
 - [10] Patricio Cortés, Luna Vattuone, José Rodriguez, Mario Duran: " A method of predictive current control with reduced number of calculations for five-phase voltage source inverters ", 2009 35th Annual Conference of IEEE Industrial Electronics, pp: 53-58, 3-5 Nov. 2009.
 - [11] Y. Zhang and X. Wei: " Torque ripple rms minimization in model predictive torque control of pmsm drives", in Electrical Machines and Systems (ICEMS), 2013 International Conference on, Oct 2013, pp:2183-2188
 - [12] R. Vargas, U. Ammann and José Rodriguez: "Predictive Approach to Increase Efficiency and Reduce Switching Losses on Matrix Converters", IEEE Trans. On Power Electronics, Vol. 24, N° 4, April 2009, pp: 894-902.
 - [13] Tobias Geyer: "Generalized Model Predictive Direct Torque Control: Long Prediction Horizons and Minimization of Switching Losses", Joint 48th IEEE Conference on Decision and Control and 28th Chinese Control Conference, Shanghai, China, December 16-18, 2009, pp: 6799-6804.
 - [14] F. Wang, Z. Chen, P. Stolze, M. Trincado, Jose Rodriguez and R. Kennel, "A comprehensive Study of Direct torque Control (DTC) and Predictive Torque Control (PTC) for High Performance Electrical Drives", 2015, EPE Journal, 25:1, 12-21
 - [15] T. Jahns, G. Kliman, T. Neumann: "Interior Permanent-Magnet Synchronous Motors for Adjustable-Speed Drives", IEEE Trans. Ind. Appl. Vol. IA-22, pp. 748-747, 1986.
 - [16] Arafat S. M., and Co.: "Comparative Study of Sensorless Control Methods of PMSM Drives", Innovative Systems Design and Engineering "IISTE", ISSN 2222-1727, vol. 2, n° 5, 2011
 - [17] C. Rojas, J. Rodriguez, F. Villarroel, J. Espinoza, C. Silva, and M. Trincado: " Predictive torque and flux control without weighting factors", Industrial Electronics, IEEE Transactions on, Vol. 60, N° 2, pp: 681-690, Feb 2013.
 - [18] F. Morel, X. Lin-Shi, J. M. Retif: "A comparative study of predictive current control schemes for a permanent-magnet synchronous machine drive ", IEEE Trans, Ind, Electron, Vol. 56, N° 7, pp: 2715-2728, July 2009.
 - [19] Jose Rodriguez Patricio Cortes "Predictive Control of Power Converters and Electrical Drives". A John Wiley & Sons, Ltd. Publication. 2012.